

What is a MOSFET?

The full form of MOSFET is metal-oxide-semiconductor field-effect transistor, it is a type of FET with an insulated gate that is assembled by the controlled oxidation of that semiconductor. The semiconductor used in it is generally silicon. In more detail, it is a three terminal-based device that is composed of a:

- Source (S)
- Gate (G)
- Drain (D)

In a general overview, the body of a MOSFET is in connection with its source terminal, hence forming a three-terminal device such as a field-effect transistor. It is normally considered a transistor and is utilized in both digital circuits and analog circuits.

MOSFET Symbol

The general configuration of a MOSFET is shown below :

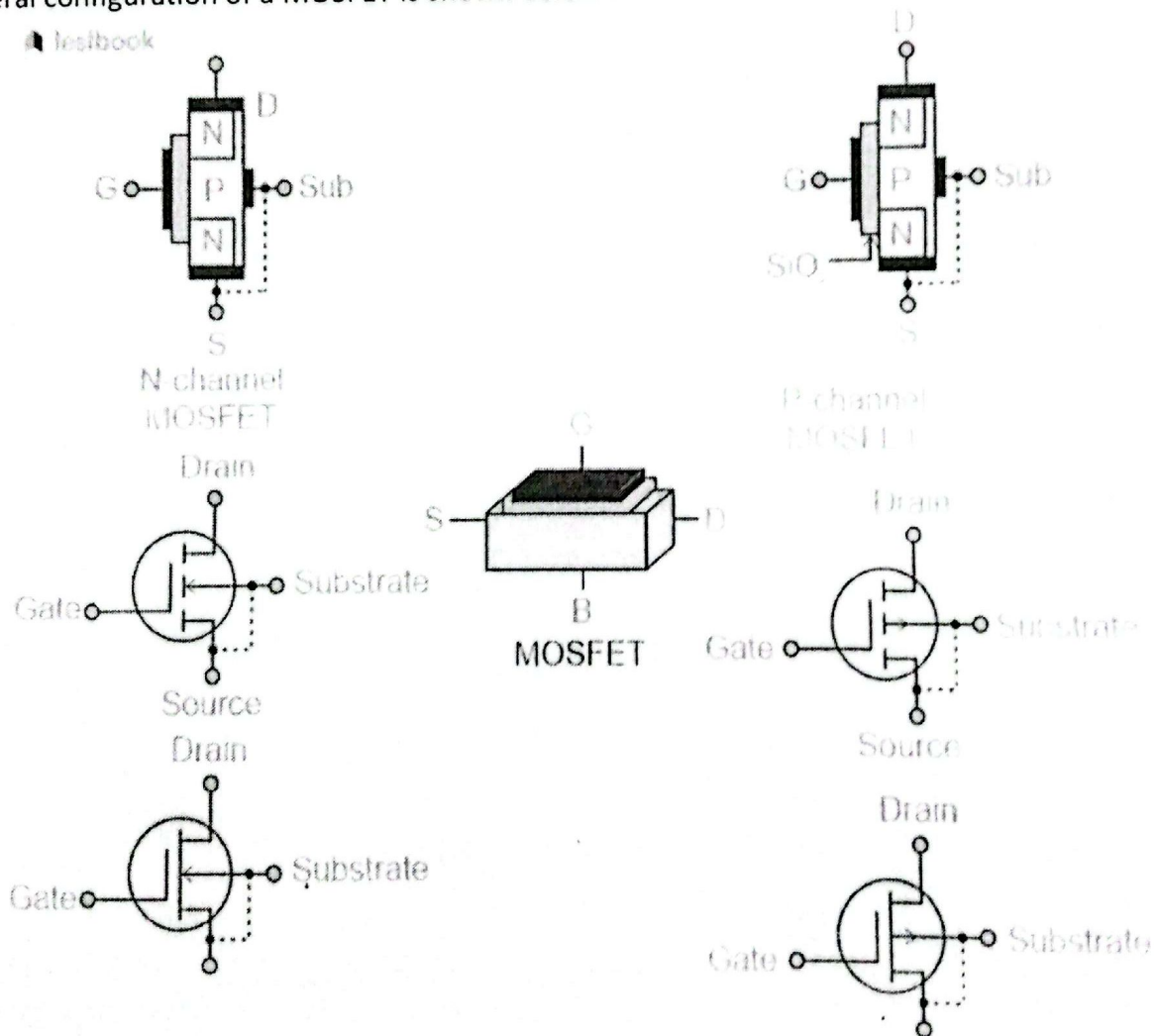


Fig-Symbol of MOSFET

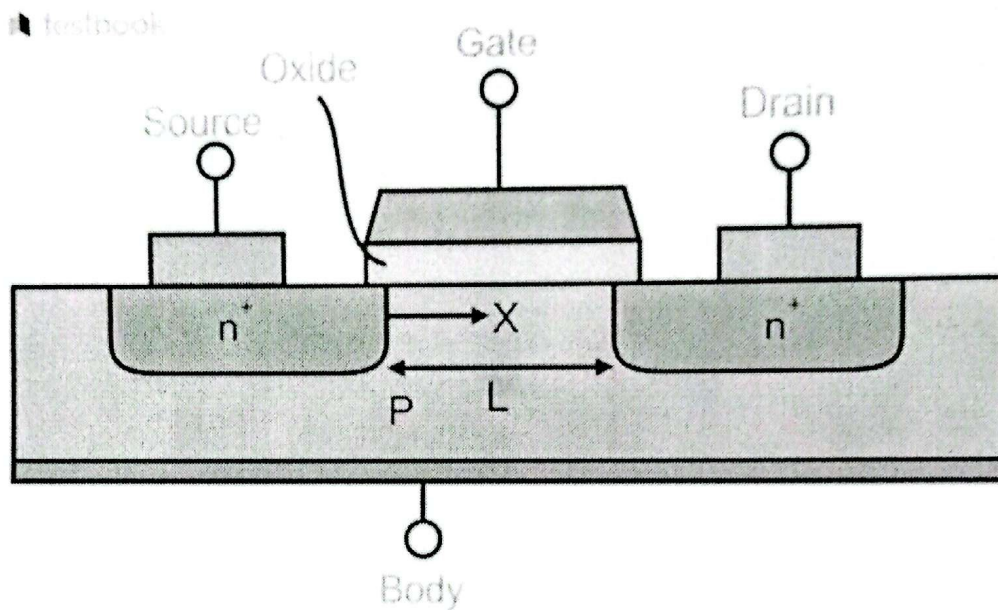
Construction of MOSFET

The circuit of a MOSFET is usually constructed as follows:

- Its base is formed of a P-type semiconductor.
- Its two types of bases are highly doped with an N-type of impurity (which is shown as n^+ in the diagram below).
- The terminals, source, and drain originate from the heavily doped region of the base.

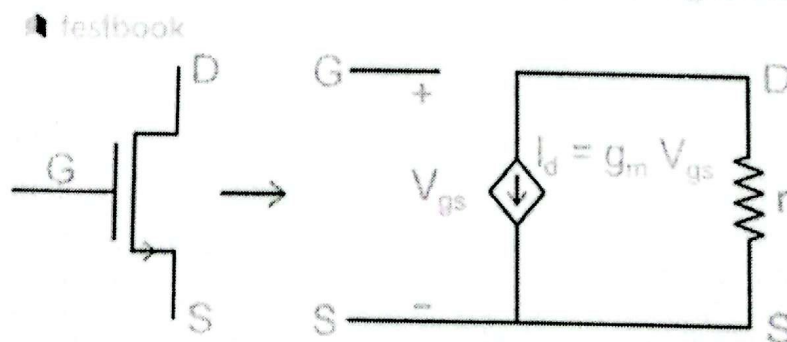
- The substrate layer is coated with a layer of insulating silicon dioxide.
- On top of silicon dioxide, a thin insulated metal plate is kept, which acts as a capacitor in the MOSFET.
- From the thin metallic plate, the gate terminal is brought out.
- A DC circuit is formed when a voltage source is connected between these two doped N-type regions.

The image below shows the construction of the MOSFET:



MOSFET Signal

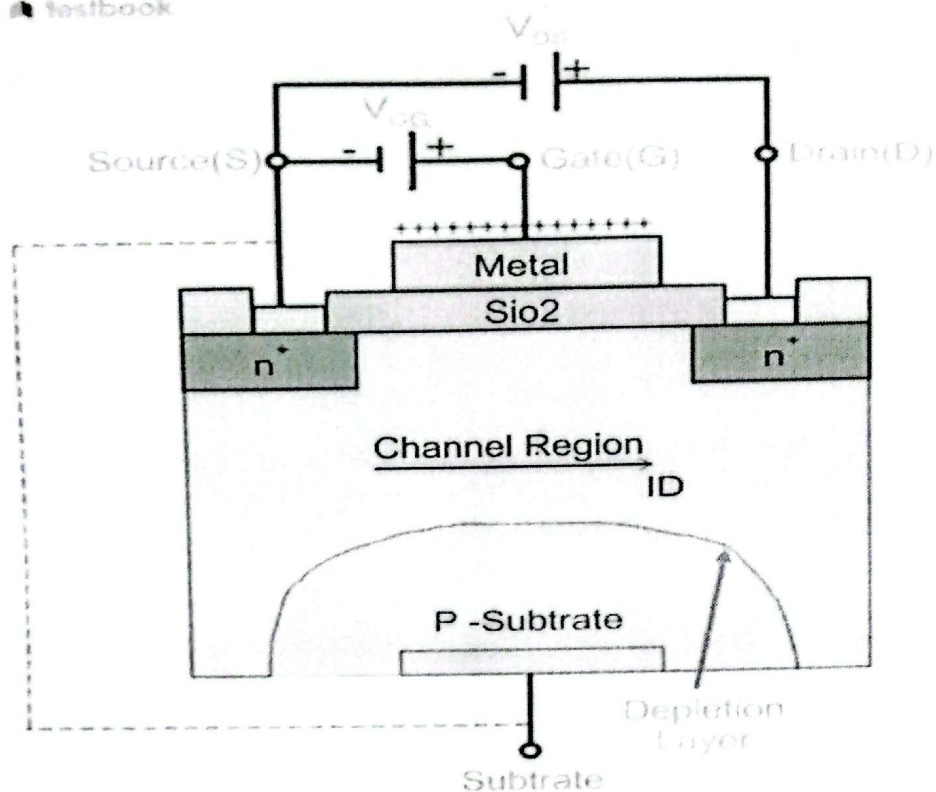
The MOSFET small-signal model works as an amplifier. Its work is mostly in the saturation region because of the huge output resistance. The small-signal model of the MOSFET is useful only as an amplifier. Its diagram is shown below to understand the idea of a small-signal model of the MOSFET.



MOSFET Small Signal Model

Working Principle of MOSFET

The main working principle of a MOSFET is to control the voltage and the current which is flowing between the source terminal and the drain terminals. We can also say that its work is almost similar to that of a switch.



The semiconductor surface which is present below the oxide layer placed between the source and drain terminal can be reversed from a P-type to an N-type by applying either a positive or negative gate of voltages respectively. When a repulsive force is applied for the positive gate voltage, then the holes present beneath the oxide layer get pushed downward with the substrate present there. The depletion region of a MOSFET is inhabited by the bounded negative charges which are linked with an acceptor atom. When electrons reach it a channel gets developed. The positive voltage also attracts electrons present in the n^+ source and drain regions towards the channel.

So, when a voltage is applied between the drain and source, the current starts to flow freely between the source and the drain. The flow of gate voltage controls the electrons present in the channel. But, if a negative voltage is applied to it, then a hole channel will start to form under the oxide layer of the MOSFET.

Types of MOSFET

- Depletion Mode
- Enhancement Mode

Enhanced Mode

Enhanced mode can be defined as if there is no voltage across the gate terminal, then the device will not conduct. But when there is a maximum voltage present across the gate terminal, then the conductivity of the device gets enhanced. These are further divided by two types based on construction:

- P Channel MOSFET
- N Channel MOSFET

Depletion Mode

Depletion mode can be defined as if there is no voltage across the gate terminal, then the device will show maximum conductance. But when the voltage across the gate terminal is either positive or

negative, then the conductivity of the device decreases. Depletion mode are have two types same like enhanced mode types, which are:

- P Channel MOSFET
- N Channel MOSFET

P-Channel MOSFET

A P-channel MOSFET is a type of MOSFET in which the semiconductor material is doped with acceptor impurities, creating an excess of positive charge carriers, or holes. When a negative voltage is applied to the gate, it repels holes from the region beneath the gate, forming a channel of positive charge carriers. This channel allows current to flow from the drain to the source.

N-Channel MOSFET

An N-channel MOSFET is a type of MOSFET in which the semiconductor material is doped with donor impurities, creating an excess of electrons. When a positive voltage is applied to the gate, it attracts electrons to the region beneath the gate, forming a channel of negative charge carriers. This channel allows current to flow from the source to the drain.

Characteristics of MOSFET

The characteristics of a MOSFET are mentioned as follows:

- It is a three terminal device.
- It is unipolar
- It is controlled by voltage
- It is a high input impedance device

Operating Regions of MOSFET

It has three operating regions, which are,

- **Cut-Off region:** This is the region in which there will be no conduction and due to which the MOSFET will stay off and will behave like an open switch.
- **Ohmic region:** This is the region where the current gets increased with an increase in its voltage value. MOSFETs used to operate in this region are generally used as amplifiers.
- **Saturation region:** This is the region that currently stays constant even when the voltage increases. This occurs only when the pinch-off value of the voltage increases. In this region the MOSFET works as a closed switch, this operating region is only used when a MOSFET is required to work in switching operations.

Applications of MOSFET

The applications of a MOSFET are mentioned below:

- **MOSFET as a Switch:** MOSFETs can be used as switches to control the flow of current. When the gate voltage is zero, the channel is not formed, and no current flows between the source and drain. When a positive voltage is applied to the gate of an N-channel MOSFET or a negative voltage is applied to the gate of a P-channel MOSFET, the channel is formed, and current flows between the source and drain.
- **MOSFET as an Amplifier:** MOSFETs can also be used as amplifiers to increase the strength of a signal. When a small voltage is applied to the gate, it can produce a large change in current between the source and drain. This property makes MOSFETs very useful in electronic circuits.
- It is used as an inverter.
- It is used in digital circuits.

- It is used as a passive element, like in an inductor, resistor, and capacitor.
- It is used as a high-frequency amplifier.
- It is used in brushless DC motor drives.
- It is used in electronic DC relays.
- It is used in SMPS.
- It is used as a switch and in amplifying electronic signals in an electronic device.

Advantages and Disadvantages of MOSFET

The advantages and disadvantages of a MOSFET are mentioned as follows:

Advantages

- It is used to work at a greater efficiency but with a lower voltage value.
- As it has no gate current so due to this it gives a high input impedance which produces a high switching speed.

Disadvantages

The disadvantages of a MOSFET are mentioned as follows:

- It is susceptible to damage due to its thin oxide layer which can get damaged because of electrostatic charges.
- It gets unstable when a voltage gets overloaded.

Difference between BJT and MOSFET

The difference between BJT and MOSFET is mentioned in the given table below:

MOSFET	BJT
1) They are mainly of two types named as ○ N-type, and ○ P-type.	1) They are also of mainly two types that are named as ○ NPN, and ○ PNP.
2) It is a device controlled by voltage.	2) It is a device controlled by current.
3) Its input resistance is high.	3) Its input resistance is low.
4) It is used in high-current applications.	4) It is used in low-current applications.